Slope, Motion & Comparing Rates

Four questions over the first two pages: solve, diamond, and two spirals.

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

Solve for x. Show each step, then check your answer.

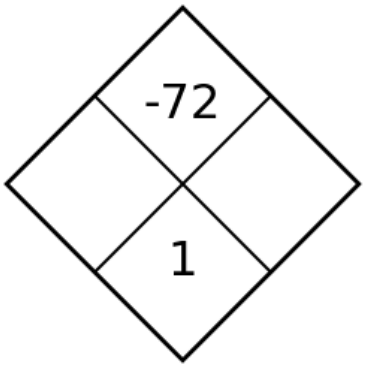
$$5(x-2)-2x=14$$

ANSWER / TEACHING NOTE

5(x - 2) - 2x = 14 → 5x - 10 - 2x = 14 → 3x = 24 → x = 8.

2 | Diamond

Find two numbers whose product is at the TOP and sum is at the BOTTOM.



SOLUTION: 9 and -8 (product -72, sum 1).

Spiral Retrieval

Retrieve ideas from earlier Algebra 1 work. Explain your reasoning.

3 | Spiral Review

A table includes (0,5), (2,11), (4,17). Find its slope.

ANSWER / TEACHING NOTE

m = (11 - 5)/(2 - 0) = 6/2 = 3.

4 | Spiral Review

Can {(1,8), (2,8), (2,9)} be a function? Explain.

ANSWER / TEACHING NOTE

No: input 2 produces two different outputs (8 and 9).

What We Need to Know

Slope, Motion & Comparing Rates

Learning focus | LG 02-ican-5 / LG 03-ican-3 / LG 03-ican-4

- I can calculate slope from two points or a table.
- I can interpret a slope using units and a direction.
- I can compare two rates fairly, including negative slopes.

Vocabulary | Highlight the important words

Rate of change

How much an output changes compared with an input change, with units attached.

Slope

The vertical change divided by the horizontal change of a straight line.

Positive / negative slope

A positive slope rises as x increases; a negative slope falls as x increases.

Zero slope

A horizontal line with no vertical change; slope zero.

Position

An object's location measured relative to a chosen reference point.

Velocity on a position-time graph

The rate and direction of change in position; the slope is signed and measured in distance per time.

Meaning & Method | Rewritten review notes

1. Always find slope with a ratio: (change in output) / (change in input), not the vertical change alone.

2. In context, read the sign and the units. A negative position-time slope can mean travel toward the chosen zero position; it does not mean negative speed.

3. To compare steepness or speed, compare magnitudes when the units are the same; explain the signs separately as direction or decreasing quantity.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Examples & You Try Its

Complete the Example with your teacher. Then solve the matching You Try It.

Example

A cart moves along a straight track. Determine the change in position per second.

Time (s)	Position (m)
0	30
2	22
4	14

Record the method and steps:

> **ANSWER / TEACHING NOTE**
> Δposition = 22 - 30 = -8 m; Δtime = 2 s; slope = -8/2 = -4 m/s. Position decreases by 4 m each second.

You Try It

Repeat with a different motion table.

Time (s)	Position (m)
0	10
3	25
6	40

Find slope with units and explain direction.

> **ANSWER / TEACHING NOTE**
> Δposition = 25 - 10 = +15 m; Δtime = 3 s; slope = +5 m/s; position increases 5 m each second.

Explain or check your approach

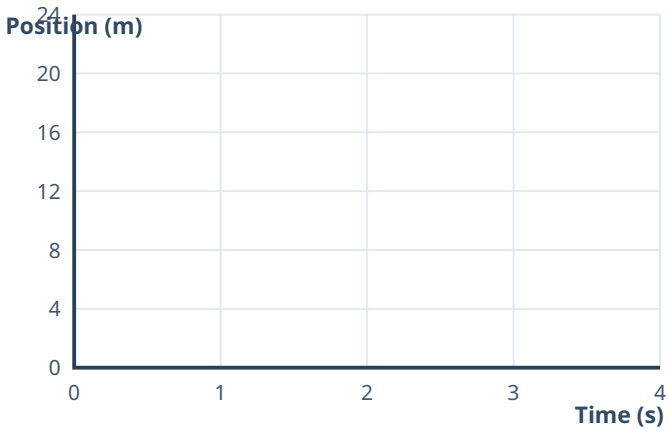
Launch | Two walkers, one coordinate system

Notice. Try it. Explain your reasoning with a partner.

The values are positions measured in meters from the same starting marker. Graph both walkers using the given data, and explain how slope indicates direction.

Time (s)	Walker A (m)	Walker B (m)
0	0	20
1	4	17
2	8	14
3	12	11

Graph key: Walker A is blue; Walker B is red.



1 | Calculate each signed slope. Which walker changes position faster in magnitude? How do you know?

ANSWER / TEACHING NOTE

Walker A: +4 m/s. Walker B: -3 m/s. Walker A has greater speed magnitude, 4 m/s.

2 | What does the negative slope for Walker B mean? Why would it be inaccurate to call Walker B's speed -3 m/s?

ANSWER / TEACHING NOTE

B moves toward smaller coordinate values; its velocity is -3 m/s but speed is 3 m/s.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Review & Preview

Independent practice. A star marks a useful quick diagnostic for the November 10 test.

1 ★ Position goes from 32 m at 0 s to 14 m at 6 s. Calculate average rate of change with units.

ANSWER / TEACHING NOTE

(14 - 32)/(6 - 0) = -18/6 = -3 m/s.

2 Find the slope between (-2,7) and (4,-5). Show Delta y and Delta x.

ANSWER / TEACHING NOTE

(-5 - 7)/(4 - (-2)) = -12/6 = -2.

3 ★ Two cyclists travel 30 m in 5 s and 42 m in 7 s. Compare their average rates.

ANSWER / TEACHING NOTE

Both are 6 m/s: 30/5 = 6 and 42/7 = 6.

Review & Preview

Independent practice. A star marks a useful quick diagnostic for the November 10 test.

4 A tank starts at 70 gallons and loses 4 gallons per minute. Interpret the starting value and rate.

ANSWER / TEACHING NOTE

Initial volume 70 gallons; change -4 gallons each minute.

5 Spiral: For {(0,5),(2,5),(2,7)}, is it a function? List domain/range.

ANSWER / TEACHING NOTE

Not a function: x=2 has two outputs. Domain {0,2}; range {5,7}.

6 Preview: A scooter is 12 meters from a marker at time 0 and its position increases by 3 meters each second. Write an equation for position p after t seconds.

ANSWER / TEACHING NOTE

p = 3t + 12 (meters, seconds).

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.